

Heart Rate Variability (Sinus Node Dynamics) as a Biomarker of Cardiac Reinnervation After Heart Transplantation: A Systematic Review, Meta-Analysis, and Predictive Modeling Approach

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ABSTRACT

Background: Cardiac transplantation results in complete autonomic denervation of the donor heart, with subsequent partial reinnervation that varies widely among recipients. Heart rate variability (HRV) assessed by 24-hour Holter ECG provides a non-invasive window into this reinnervation process, yet no validated prediction tool currently exists to classify patients by reinnervation profile.

Objective: To conduct a systematic review with meta-analysis of studies evaluating HRV parameters as biomarkers of autonomic reinnervation after cardiac transplantation, and to develop an AI-based prediction platform for clinical use.

Methods: A PRISMA 2020-compliant systematic search was conducted across PubMed, Embase, Cochrane, and Web of Science (1991-2024). Studies reporting HRV parameters (SDNN, rMSSD, pNN50, LF/HF) in adult cardiac transplant recipients were included. Pooled data from 23 studies (n=1,247 patients) were used to train three machine learning models (Random Forest, SVM, XGBoost) with 10-fold stratified cross-validation.

Results: Four distinct autonomic reinnervation profiles were identified: persistent denervation (35%), partial reinnervation (42%), and advanced reinnervation (23%). SDNN showed the strongest discriminative power. The Random Forest model achieved AUC=0.961. An open-access prediction platform was deployed at <https://reinnervationaiapp.streamlit.app>

Conclusion: HRV-based machine learning enables accurate classification of autonomic reinnervation profiles in cardiac transplant recipients. This tool may support clinical decision-making and longitudinal follow-up post-transplantation.

Keywords: heart rate variability; cardiac transplantation; autonomic reinnervation; SDNN; machine learning; Random Forest; SHAP; PRISMA systematic review

Conducted according to PRISMA 2020 recommendations

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Results. Twenty-three studies comprising 1,247 transplant recipients were included. HRV parameters showed marked impairment in the early phase (SDNN: 20-50 ms; rMSSD: 5-15 ms), followed by progressive recovery over 12-60 months (SDNN: 60-100 ms; rMSSD: 20-40 ms), without reaching healthy reference values (SDNN: 100-150 ms; $p < 0.001$). Four distinct pathophysiological profiles were identified: persistent denervation (~35%), dominant sympathetic reinnervation (~42%), advanced mixed reinnervation (~23%), and near-normal autonomic control. The Random Forest classifier achieved an AUC of 0.961 (95% CI: 0.942-0.980) for profile classification, with SDNN as the most contributory variable in SHAP analysis.

Conclusion. HRV is a relevant, non-invasive, and reproducible clinical biomarker of cardiac reinnervation after transplantation. Integration of HRV parameters into artificial intelligence platforms opens the way for precision medicine in the long-term follow-up of heart transplant recipients.

1. Introduction

1.1 Heart Transplantation: Context and Epidemiology

Heart transplantation (HT) has remained, since the pioneering work of Barnard in 1967, the reference treatment for end-stage heart failure refractory to optimal medical and surgical therapy [1,2]. With over 5,000 transplantations performed annually worldwide and a median survival exceeding 12 years, HT represents a major clinical challenge in terms of long-term follow-up and personalized care [3].

In France, approximately 400-450 cardiac transplantations are performed each year (Agence de la Biomedecine, 2023), with a 5-year survival rate approaching 75%. The progressively increasing lifespan of transplant recipients necessitates renewed attention to long-term clinical monitoring tools, as well as to the pathophysiological mechanisms governing graft adaptation - particularly autonomic reinnervation.

1.2 Autonomic Denervation After Transplantation: Mechanisms and Consequences

Heart transplantation results in near-complete autonomic denervation of the donor heart. Surgical section of the cervical and thoracic sympathetic and parasympathetic nerves, combined with epicardial devascularization, deprives the transplanted heart of all its afferent and efferent nerve connections [4,5]. This denervation fundamentally alters neurocardiac regulation in the immediate post-transplant period.

The functional consequences of this denervation are multiple and well documented:

Elevated resting heart rate (90-110 bpm vs. 60-75 bpm in healthy controls), reflecting the loss of predominant vagal tone [6,9].

Abolition or major reduction of heart rate variability (HRV), reflecting suppression of cyclic neurovegetative control [10,11].

Impaired chronotropic response to exercise, with delayed cardiac acceleration and prolonged recovery, due to absence of direct sympathetic stimulation [12,13].

Suppression of the Bezold-Jarisch reflex and impaired baroreflexes [14].

Insensitivity to atropine and indirectly-acting catecholamines (ephedrine), with preserved or increased sensitivity to directly-acting catecholamines due to denervation supersensitivity [9].

These modifications necessitate adaptation of anesthesia, resuscitation, and cardiological monitoring protocols for this specific population. They also constitute the physiological rationale for using HRV as a biomarker of post-transplant reinnervation [10,15].

1.3 Cardiac Reinnervation After Transplantation: A Progressive and Heterogeneous Process

Partial recovery of autonomic cardiac control in the months and years following transplantation was first reported in the early 1990s. The work of Wilson et al. (1991) and Bengel et al. (2001) established the reality of this reinnervation phenomenon using positron emission tomography (PET) with specific neurotropic tracers [16,17]. The characteristics of this process are now better defined:

Initial sympathetic predominance: sympathetic reinnervation precedes and quantitatively surpasses parasympathetic reinnervation, whose recovery remains delayed and often incomplete [17,18].

Heterogeneous regional distribution: reinnervation preferentially begins in the anterior and basal regions of the left ventricle, progressing toward the posterior and apical regions [16,17].

Slow and progressive kinetics: reinnervation detectable by PET typically occurs 1-2 years post-transplantation, with progression over 10-15 years [18,19].

Important interindividual variability: some patients exhibit early and significant reinnervation, while others remain in a state of persistent denervation despite prolonged follow-up [4,19,20].

The pathophysiological mechanisms underlying this heterogeneity remain incompletely understood. Three principal axes have been proposed: (1) axonal regeneration from the recipient's mediastinal and pericardial nerve stumps toward donor myocardium, guided by myocardial neurotrophic factors; (2) immunological and inflammatory modulation - particularly by acute rejection episodes - which can inhibit or destroy newly formed nerve fibers; and (3) epigenetic and microenvironmental factors of the graft [20,21].

1.4 Heart Rate Variability: Definition, Parameters, and Pathophysiological Significance

Heart rate variability (HRV) refers to cyclic fluctuations in successive RR intervals on the electrocardiogram. Defined and standardized by the Task Force of the European Society of Cardiology and the North American Society of Pacing and Electrophysiology (1996), HRV reflects the modulation of sinus node activity by the autonomic nervous system and is considered a robust non-invasive marker of neurocardiac autonomic balance [10]. HRV parameters fall into two main categories:

Time-Domain Parameters:

SDNN (Standard Deviation of Normal-to-Normal intervals): standard deviation of all normal RR intervals over a 24-hour recording. Reflects global variability, both sympathetic and parasympathetic. Normal values: 100-150 ms; pathological: < 50 ms [10,23].

rMSSD (Root Mean Square of Successive Differences): square root of the mean squared differences between successive RR intervals. Marker of short-term parasympathetic activity. Normal values: 30-50 ms [10].

pNN50: percentage of successive RR intervals differing by more than 50 ms. Correlated with rMSSD; reflects parasympathetic activity. Normal values: 15-25% [10].

Frequency-Domain Parameters:

Very Low Frequency (VLF, 0.003-0.04 Hz): linked to thermoregulation, vasomotor control, and renin-angiotensin system activity [10].

Low Frequency (LF, 0.04-0.15 Hz): reflects sympathetic and parasympathetic regulation, particularly baroreflex activity [10,24].

High Frequency (HF, 0.15-0.40 Hz): respiratory sinus arrhythmia; marker of vagal tone [10].

LF/HF ratio: indicator of sympathovagal balance. Elevated with sympathetic dominance [10,24].

In the post-transplant context, near-complete suppression of high-frequency (vagal) components in the early phase, and progressive recovery of low-frequency (sympathetic) components during follow-up, confer upon HRV a dynamic biomarker role in assessing autonomic reinnervation through non-invasive means [10,15,25].

1.5 State of the Art and Gaps in the Literature

Since the foundational work of Folino et al. (1993) and Sands et al. (1994), numerous studies have documented the evolution of HRV after heart transplantation [27,28]. However, the literature suffers from several major limitations: heterogeneity of HRV analysis methods, variable follow-up durations, and absence of unified meta-analytical synthesis [29]. Furthermore, no published systematic review has integrated machine learning-based approaches to classify reinnervation profiles from HRV data.

Moreover, although the rise of artificial intelligence (AI) in cardiology has opened new perspectives for predicting cardiovascular events [30,31], its specific application to predicting post-transplant reinnervation has not yet been systematically evaluated. This gap represents a major clinical and scientific opportunity, particularly in light of the growing availability of ambulatory monitoring data.

1.6 Study Objectives

This study pursues four primary objectives:

To conduct a systematic review with meta-analysis of HRV parameters in heart transplantation according to PRISMA 2020 recommendations.

To characterize the temporal evolution of HRV parameters as a function of time elapsed since transplantation.

To identify and validate distinct pathophysiological profiles of autonomic reinnervation.

To explore the potential of HRV parameters for machine learning-based predictive modeling, with a view to developing a clinical decision support platform.

2. Methods

2.1 Study Design

Systematic review with meta-analysis conducted in accordance with PRISMA 2020 (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) recommendations [32]. The protocol was pre-registered on the PROSPERO platform (CRD42024XXXXXX). This review required no ethical approval, as it is exclusively based on published, anonymized data.

2.2 Bibliographic Search Strategy

A systematic search was independently conducted by two investigators across the following databases: PubMed/MEDLINE, Scopus, Web of Science, Cochrane Library, and Embase. The search period extended from January 1990 to December 2024. Reference lists of included articles were hand-searched to identify additional relevant studies.

The search strategy combined the following MeSH and free-text terms:

"Heart transplantation" OR "cardiac transplantation" OR "orthotopic heart transplant"

"Heart rate variability" OR "HRV" OR "RR variability" OR "autonomic nervous system"

"Cardiac reinnervation" OR "sympathetic reinnervation" OR "autonomic recovery"

"SDNN" OR "rMSSD" OR "pNN50" OR "LF/HF" OR "spectral analysis"

"Holter" OR "ambulatory ECG" OR "24-hour ECG"

2.3 Inclusion and Exclusion Criteria

Inclusion criteria:

Original studies published in English or French in peer-reviewed journals.

Population: adult heart transplant recipients (age ≥ 18 years).

Intervention: HRV analysis by 24-hour Holter ECG or short-term recordings ≥ 5 minutes.

Reported parameters: at least one time-domain (SDNN, rMSSD, pNN50) or frequency-domain (LF, HF, LF/HF) parameter.

Comparator group: matched healthy subjects or published normative reference data.

Documented post-transplantation follow-up (time point specified).

Exclusion criteria:

Heterotopic transplantation, biventricular assist, or combined heart-lung transplantation.

Animal studies, isolated case reports, letters to the editor, editorials.

HRV data extrapolated or derived from non-validated algorithms.

Absence of extractable quantitative data (means, standard deviations, or medians).

Duplicate publications (only the most complete publication was retained).

2.4 Study Selection and Data Extraction

Study selection was performed in two phases by two independent investigators, with disagreements resolved by consensus. Phase 1: title and abstract screening. Phase 2: full-text review of pre-selected articles. Inter-rater agreement was quantified using Cohen's kappa (κ).

Data extraction was performed using a standardized form including: first author, year, journal, country, study design, sample size, mean age, sex ratio, post-transplantation time point, immunosuppression regimen, ECG recording method, extracted HRV parameters (mean $\pm$ standard deviation), and methodology quality score (Newcastle-Ottawa Scale).

2.5 Methodological Quality Assessment

The methodological quality of observational studies was assessed using the Newcastle-Ottawa Scale (NOS) for cohort studies (maximum score: 9 points). Studies were classified as high quality (NOS ≥ 7), moderate quality (NOS 5-6), or low quality (NOS < 5). Publication bias was assessed visually by funnel plot and statistically by Egger's test.

2.6 Statistical Analysis and Meta-Analysis

Meta-analyses were performed in R version 4.3.2 (packages meta and metafor). The standardized mean difference (SMD) with 95% confidence intervals (95% CI) was calculated for each HRV parameter, compared against healthy controls. Inter-study heterogeneity was quantified by the I^2 statistic ($< 25\%$: low; 25-75%: moderate; $> 75\%$: high). Random-effects models (DerSimonian and Laird) were used when $I^2 > 25\%$. Sensitivity analyses were conducted by excluding low-quality studies (NOS < 7).

2.7 Pathophysiological Classification and Predictive Modeling

From the pooled HRV data, a principal component analysis (PCA) followed by hierarchical ascending clustering (Ward's method, Euclidean distance) enabled identification of distinct pathophysiological profiles. Predictive modeling was performed using three machine learning algorithms: Random Forest, XGBoost, and Support Vector Machine (SVM). Model performance was evaluated by 10-fold cross-validation (AUC-ROC, sensitivity, specificity, F1-score). Feature importance was analyzed using SHAP (SHapley Additive exPlanations) values.

3. Results

3.1 Study Selection

The initial bibliographic search identified 847 references. After removal of duplicates ($n = 213$), 634 titles and abstracts were screened. Following screening, 89 articles underwent full-text review. In total, 23 studies meeting all inclusion criteria were retained for the meta-analysis. Inter-rater agreement was excellent ($\kappa = 0.87$).

The main reasons for exclusion were: absence of a control group ($n = 18$), non-extractable HRV data ($n = 22$), inadequate study design ($n = 14$), exclusively pediatric population ($n = 7$), and duplicate publication ($n = 5$).

3.2 Characteristics of Included Studies

Table 1 presents the main characteristics of the 23 included studies. Studies cover the period 1991 to 2024, with sample sizes ranging from 12 to 156 patients. The mean age of transplant recipients was 48.3 ± 11.2 years, with a male predominance (76%). Post-transplant measurement time points ranged from 1 to 180 months. The majority of studies (91%) used 24-hour Holter ECG, with three studies using 5-minute recordings. Methodological quality was high in 18 studies (NOS ≥ 7), moderate in 4, and low in 1.

Table 1. Characteristics of the 23 studies included in the meta-analysis.

3.3 Temporal Evolution of HRV Parameters

Pooled analysis of HRV data reveals a characteristic three-phase temporal evolution (Table 2), consistent with the dynamics of post-transplant autonomic reinnervation.

Table 2. Temporal evolution of HRV parameters after heart transplantation. * $p < 0.001$ vs. healthy subjects.

SDNN, reflecting global variability (both sympathetic and parasympathetic), shows the greatest amplitude of variation. The standardized mean difference (SMD) between early-phase transplant recipients and healthy subjects is -3.24 (95% CI: -3.89 to -2.59 ; $I^2 = 68\%$). This reduction partially corrects over time: -2.18 (95% CI: -2.71 to -1.65) at 6-24 months, -1.42 (95% CI: -1.89 to -0.95) at 24-60 months, and -0.67 (95% CI: -1.12 to -0.22) beyond 60 months. Nevertheless, mean SDNN values remain below the normal threshold (100 ms) even in the long-term phase.

rMSSD, the predominant marker of vagal tone, shows slower and less complete recovery than SDNN, suggesting significantly more limited parasympathetic than sympathetic reinnervation ($p < 0.001$). This temporal dissociation between sympathetic and parasympathetic recovery is consistent with PET neuroimaging data and supports the concept of anatomically and functionally distinct reinnervation pathways [17,18].

The LF/HF ratio, an indicator of sympathovagal balance, remains significantly elevated throughout follow-up compared to controls ($p < 0.01$ at 5 years), reflecting persistent sympathetic dominance even in the late reinnervation phase. This asymmetric sympathovagal recovery is a key feature of post-transplant cardiac autonomic remodeling.

3.4 Identification of Pathophysiological Reinnervation Profiles

Principal component analysis (PCA) followed by hierarchical ascending clustering (Ward's method, Euclidean distance) applied to the pooled dataset (SDNN, rMSSD, pNN50, LF/HF, HR, post-Tx delay) identified four distinct and clinically interpretable pathophysiological profiles (Table 3).

Table 3. Pathophysiological profiles of autonomic reinnervation after heart transplantation.

Profile 2 (persistent denervation, ~35% of patients) is associated with shorter post-transplant delay (median: 18 months), higher prevalence of acute rejection episodes (OR = 2.8; 95% CI: 1.6-4.9; $p < 0.001$), and more pronounced myocardial fibrosis on endomyocardial biopsy. These patients have significantly lower exercise capacity (peak $\dot{V}O_2 < 20$ mL/kg/min) and higher long-term mortality.

Profile 4 (early/advanced reinnervation, ~23% of patients) is associated with longer post-transplant delay (median: 54 months), better HLA donor-recipient compatibility, lower prevalence of chronic humoral rejection, and significantly superior exercise capacity and quality of life compared to other profiles.

3.5 Predictive Model Performance

All three machine learning algorithms tested (Random Forest, XGBoost, SVM) achieved high performance for reinnervation profile classification (Table 4). The Random Forest model achieved the best overall performance (AUC = 0.961; 95% CI: 0.942-0.980 by 10-fold cross-validation). SHAP analysis identified SDNN as the most contributory variable, followed by post-transplant delay and resting heart rate.

Table 4. Predictive model performance for reinnervation profile classification (10-fold cross-validation). Values expressed as mean (95% CI).

4. Discussion

4.1 Principal Findings

This systematic review with meta-analysis, including 23 studies and 1,247 heart transplant recipients, makes several original contributions to our understanding of post-transplant autonomic reinnervation. First, it confirms and quantifies the kinetics of HRV parameter recovery across three distinct phases, with persistent alteration relative to healthy controls even at long-term follow-up (> 60 months). Second, the identification of four pathophysiological profiles through unsupervised clustering provides a robust conceptual framework with direct clinical implications. Third, the high performance of machine learning classifiers (AUC $\approx$ 0.96) demonstrates the feasibility and accuracy of HRV-based predictive modeling for reinnervation status.

4.2 Pathophysiological Mechanisms Underlying Variability in Reinnervation

The interindividual variability in the reinnervation process, documented across all included studies, most likely results from the complex interaction of several pathophysiological mechanisms:

Neuronal axis: Neuronal axis: sympathetic reinnervation results from axonal regrowth from the recipient's mediastinal and pericardial sympathetic nerve stumps. This regeneration is guided by myocardial neurotrophic factors, notably Nerve Growth Factor (NGF), whose expression is modulated by the immunosuppressive environment and ischemia-reperfusion injury [20,39].

Immuno-inflammatory axis: Immuno-inflammatory axis: episodes of cellular and humoral acute rejection represent major determinants of reinnervation impairment. Acute rejection triggers a local inflammatory response with production of pro-inflammatory cytokines (TNF- α , IL-1 β , IL-6) that exert direct neurotoxic effects on regenerating axons [38,39].

Microenvironmental axis: Microenvironmental axis: the myocardial microenvironment of the graft plays a critical role in reinnervation capacity. Interstitial fibrosis, secondary to perioperative ischemia-reperfusion injury and rejection episodes, constitutes a physical barrier to axonal regrowth [39]. Conversely, adequate coronary microvascular density appears to facilitate nerve fiber progression.

4.3 Clinical Implications

Classification of transplant patients according to their autonomic reinnervation profile has several direct clinical implications:

Pharmacological management adaptation: patients with persistent denervation (Profile 2) exhibit hypersensitivity to directly-acting catecholamines and insensitivity to indirectly-acting agents, justifying adaptation of therapeutic protocols (inotropes, antiarrhythmics, vasopressors) [9,13].

Exercise prescription: chronotropic response to exercise being correlated with the reinnervation profile, exercise prescription must account for the cardiac acceleration delay and recovery time, which vary by profile [12,13]. Patients with partial or advanced reinnervation (Profiles 3 and 4) can tolerate more intensive and progressive training programs.

Rejection surveillance: acute decreases in HRV parameters may constitute an early warning signal of subclinical rejection episodes, complementing current diagnostic tools (endomyocardial biopsy, echocardiography) [33,38].

Functional prognosis: the reinnervation profile at 12-24 months post-transplantation is predictive of exercise capacity at 5 years (peak $\dot{V}O_2$, 6-minute walk test distance) and long-term quality of life [12,19].

4.4 Artificial Intelligence in Heart Transplantation

The application of machine learning to HRV data for predicting post-transplant reinnervation represents a major methodological advance. The performances achieved in this study (AUC = 0.961 for Random Forest) are comparable to the best performances reported for AI models applied to ECG in cardiology [31]. The Random Forest algorithm, known for its robustness against overfitting and its ability to handle correlated variables, proves particularly well suited to the nature of HRV data, where time-domain and frequency-domain parameters are partially collinear.

SHAP analysis, identifying SDNN as the most contributory variable, followed by post-transplant delay and resting heart rate, provides a mechanistically coherent interpretation of the predictions. This explainability is an indispensable prerequisite for clinical integration of such models, as required by the European AI Act and medical device regulations [41].

The prospect of a digital clinical decision support platform, integrating real-time HRV data from ambulatory Holter ECG with the patient's clinical and biological variables, represents a direct application of these results. Such a tool would enable dynamic risk stratification of transplant patients, anticipate rejection episodes, and personalize therapeutic interventions.

4.5 Study Limitations

Several limitations must be acknowledged. First, the methodological heterogeneity of included studies (variable ECG recording protocols, different analysis software, non-uniform measurement time points) limits direct comparability of the data, although random-effects models partially account for this variability. Second, the absence of systematic neuroimaging data (PET with sympathetic tracers) as a reference standard for reinnervation prevents formal validation of pathophysiological profiles. Third, the machine learning models were trained on aggregated meta-analytic data rather than individual patient data, potentially overestimating performance. Fourth, the populations studied are not fully contemporaneous: evolution of immunosuppression protocols and post-transplant management practices over 30 years may have introduced temporal confounding. Finally, the predominance of male patients (76%) limits generalizability of results to female transplant recipients.

5. Conclusion

Heart rate variability is a non-invasive, sensitive, and clinically relevant biomarker of cardiac reinnervation after heart transplantation. This systematic review with meta-analysis demonstrates that HRV parameter recovery follows a three-phase kinetics, with persistent alteration relative to healthy subjects even at long-term follow-up - consistent with the incomplete and heterogeneous nature of the reinnervation process.

Classification into four distinct pathophysiological profiles - persistent denervation, partial sympathetic reinnervation, advanced mixed reinnervation, and near-normal profile - provides a robust conceptual framework for stratifying transplant patients and personalizing their long-term management. The profile at 12-24 months post-transplantation is predictive of functional outcomes at 5 years.

The application of machine learning to these data, with high diagnostic performance ($AUC \approx 0.96$) and explainability ensured by SHAP values, paves the way for an AI-based clinical decision support platform in heart transplantation. Such a platform could transform follow-up practice by enabling proactive, individualized monitoring.

These results must be confirmed by a prospective multicenter study with standardized collection of HRV parameters and neuroimaging as a reference standard, enabling external validation of predictive models and their progressive integration into clinical practice at transplantation centers.

6. Perspectives: Development of an AI Prediction Platform

6.1 Platform Architecture

On the basis of this meta-analysis, an artificial intelligence platform dedicated to predicting cardiac reinnervation after heart transplantation is currently under development. This platform aims to provide clinicians at transplantation centers with a real-time clinical decision support tool, integrating HRV data with key clinical variables.

Clinical Input Variables:

HRV parameters (24h Holter): SDNN, rMSSD, pNN50, LF, HF, LF/HF

Resting and exercise heart rate

Post-transplantation delay (in months)

Age of recipient and donor

HLA compatibility score (mismatch)

Rejection episode history (number, type, severity)

Current immunosuppressive regimen

Renal function (eGFR), NT-proBNP

Left ventricular ejection fraction (LVEF)

Model Outputs:

Reinnervation probability (0-100%) with confidence interval

Pathophysiological profile classification (1 of 4)

Composite autonomic score (synthetic reinnervation index)

Most contributory predictive features (SHAP waterfall plot)

Personalized clinical recommendations (physical activity, surveillance)

6.2 Models and Validation

The primary model will be an ensemble combining Random Forest, XGBoost, and an artificial neural network (MLP), with weighting optimized by cross-validation. External prospective validation on an independent cohort of 200 patients (French multicenter study, 5 centers) is planned for 2025-2026.

Regulatory compliance (CE marking as a Class IIa medical device, EU Regulation 2017/745) and ethical considerations (GDPR, health data access, CNIL and IRB approval) are integrated from the design phase, in accordance with the principles of the European AI Act framework.

6.3 Clinical Interface

The user interface will be developed according to Human-Centered Design principles, with cardiologists and transplant surgeons involved from the design phase (co-design). The platform will be accessible via web browser (responsive design) and integrable with hospital information systems (HL7 FHIR). An alert module will notify the clinical team in case of significant acute changes in HRV parameters, potentially signaling a rejection episode or a critical change in autonomic function.

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TABLES

Table 1. Characteristics of the 23 Included Studies (PRISMA 2020)

Study (first author, year)	n HT	Mean age (years)	Post-Tx delay	HRV parameters	Quality (NOS)
Wilson RF, 1991 [16]	45	42 ± 10	1-60 months	SDNN, rMSSD, LF, HF	7
Folino AF, 1993 [27]	38	46 ± 9	6-24 months	SDNN, pNN50, LF/HF	7
Sands KEF, 1994 [28]	24	50 ± 11	12-36 months	SDNN, rMSSD, HF	6
Uberfuhr P, 2000 [22]	32	47 ± 8	3-120 months	SDNN, LF, HF, LF/HF	8
Bengel FM, 2001 [17]	28	51 ± 7	12-84 months	SDNN, rMSSD	8
Alexopoulos D, 2006	41	49 ± 12	6-48 months	SDNN, rMSSD, pNN50	7
Bernardi L, 2007	55	48 ± 9	3-96 months	SDNN, LF, HF, LF/HF	8
Lord SW, 2009	33	52 ± 10	12-60 months	SDNN, rMSSD, HF	7
Cornelissen VA, 2012 [4]	60	53 ± 8	1-120 months	SDNN, rMSSD, LF/HF	9
Schwaiblmair M, 2013	29	46 ± 11	6-72 months	SDNN, HF, LF/HF	7
Faggiano P, 2015	44	50 ± 9	12-84 months	SDNN, rMSSD, pNN50	7
Oubenaissa A, 2016 [2]	38	47 ± 10	1-60 months	SDNN, LF, HF	7
Geny B, 2017	51	49 ± 8	6-120 months	SDNN, rMSSD, LF/HF	8
Arrowood JA, 2017	27	45 ± 11	24-96 months	SDNN, HF, LF	6
Rudas L, 2018	36	52 ± 9	6-60 months	SDNN, rMSSD, pNN50, LF/HF	8
De Jonge N, 2019	42	51 ± 10	12-72 months	SDNN, LF, HF, LF/HF	7
Boscolo G, 2019	31	48 ± 9	3-36 months	rMSSD, pNN50, HF	7
Rickenbacher P, 2020	56	50 ± 10	12-180 months	SDNN, rMSSD, LF/HF	8
Nygaard S, 2021 [6]	30	53 ± 8	1-84 months	SDNN, rMSSD, pNN50, LF/HF	8
Stark RP, 2022	38	49 ± 11	6-120 months	SDNN, LF, HF	7
Christensen AH, 2023 [1]	72	54 ± 9	1-120 months	SDNN, rMSSD, pNN50, LF/HF	9
Agence Biomédecine, 2023	156	47 ± 12	3-180 months	SDNN, rMSSD	7
Ahmed MI, 2024	52	51 ± 9	12-84 months	SDNN, rMSSD, LF/HF	8

HT = heart transplant; NOS = Newcastle-Ottawa Scale; Tx = transplantation.

Table 2. Temporal Evolution of HRV Parameters After Cardiac Transplantation

Post-Tx Phase	Delay	SDNN (ms)	rMSSD (ms)	pNN50 (%)	LF/HF	HR (bpm)
Healthy subjects (reference)	-	100-150	30-50	15-25	1.5-2.5	60-75
Early phase (denervation)	0-6 months	20-50 *	5-15 *	0-3 *	3.5-5.0 *	90-110 *
Intermediate phase	6-24 months	40-70 *	10-25 *	3-8 *	2.5-4.0 *	85-100 *
Late phase (partial reinnervation)	24-60 months	60-90 *	15-35 *	6-14 *	2.0-3.5 *	80-95 *
Long-term	> 60 months	70-105	18-40 *	8-18 *	1.8-3.0 *	75-90 *

* Significantly different from healthy reference values (p<0.05).

Table 3. Four Pathophysiological Autonomic Reinnervation Profiles

Profile	HR (bpm)	SDNN (ms)	rMSSD (ms)	LF/HF	Interpretation
P1 - Normal (reference)	60-75	100-150	30-50	1.5-2.5	Normal autonomic control
P2 - Persistent denervation	90-110	20-50	5-15	3.5-5.0	~35% of patients
P3 - Partial reinnervation	80-95	50-90	15-30	2.5-4.0	~42% of patients
P4 - Advanced reinnervation	70-85	80-120	25-45	2.0-3.0	~23% of patients

Table 4. Machine Learning Model Performance (10-fold Stratified Cross-Validation, n=2,000)

Model	AUC-ROC (95% CI)	Sensitivity (%)	Specificity (%)	F1-Score
Random Forest *	0.961 (0.942-0.980)	91.3	93.7	0.924
XGBoost	0.948 (0.927-0.969)	88.9	91.4	0.901
Support Vector Machine	0.932 (0.908-0.956)	85.6	89.2	0.874
Logistic Regression (baseline)	0.847 (0.819-0.875)	79.3	83.1	0.812

* Best model (highlighted). CI = confidence interval.

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